

YULIA VOLKOVA

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I am a MATS 9.1 research fellow working on AI control evaluations. I have 8 years of experience building insider-threat detection and communication-surveillance systems at Behavox for Tier-1 financial institutions. I witnessed firsthand the integration of AI into the regulatory sector, which inspired me to move into AI safety. Further details can be found on my [website](#).

SELECTED PUBLICATIONS

2026	Control Evaluations for Covert Malicious Fine-Tuning Buckworth, T., Moreno, R., Panda, A., Volkova, Y. Supervised by Dr Francis Rhys Ward.	<i>Submitted to NeurIPS 2026</i>
2026	Commitment to Cooperation With Self-Negotiated Contracts Wyse, T., Bustos, K., Volkova, Y. , Kleiman-Weiner, M. Algoverse AI Safety Research Fellowship, Cooperative AI track. Mentor: Max Kleiman-Weiner.	<i>COLM 2026</i>
2025	ParaScopes: What Do Language Model Activations Encode About Future Text? Pochinkov, N., Volkova, Y. , Vasileva, A., Chereddy, S. V. R. Algoverse AI Safety Research Fellowship. PI: Callum McDougall; mentor: Nicky P.	

RESEARCH PROGRAMS

Jan 2026–Present	MATS 9.0 - Control evaluations for covert malicious fine-tuning under Francis Rhys Ward, with Tyler Tracy and James Lucassen acting as advisors - Selected for the MATS 9.0 Spotlight Talk
Nov 2025	Athena Women in AI Safety Research Fellowship Mechanistic interpretability approach to monitoring chain-of-thought faithfulness under Arun Jose
Oct 2025	SPAR Fellowship Chain-of-thought monitorability project under Qiyao Wei, MATS 8.0 and Cambridge University
Aug 2025	Algoverse AI Safety Research Fellowship - Mechanistic interpretability under Nicky P. - Cooperative AI under Max Kleiman-Weiner
Jul 2025	ARBOx2 Oxford AI Safety Bootcamp Completed the ARENA curriculum

WORK EXPERIENCE

Aug 2017–Sep 2025	Behavox Ltd. <i>AI-powered compliance and conduct surveillance for financial institutions</i>
Jul 2020–Sep 2025	Senior Data Scientist, ML Development Key responsibilities - Led the development of Insider Threat ML applications from the Data Science team perspective. These applications analyzed workplace text communications and flagged insider-threat indicators for investigation - Oversaw and took full responsibility for bringing ML applications into production - Designed ML applications and maintained end-to-end ownership of their production ML pipelines - Managed technical implementation, including developing Java backend and Groovy components to integrate ML applications into the production system - Directed and oversaw all ongoing production tasks related to ML applications

Selected technical achievements

- Ran RoBERTa sequence-length experiments that preserved model performance while increasing training throughput by 2× and reducing compute costs by 50%
- Developed a synthetic-data generation pipeline that automatically created and uploaded test datasets to AWS S3 through Jenkins, modernizing the QA process
- Created an automated process that reduced ML application development time from 3 days to 10 minutes
- Consolidated 4 legacy security-monitoring products into 1 scalable solution, reducing backend maintenance requirements

Aug 2017–Jun 2020

ML Applications Developer

Key responsibilities: development of multilingual applications and reports to flag statistical deviations and compliance breaches in client data, designing and building infrastructure to support application creation, QA and maintenance.

- Landed proof of concepts that resulted in 3+ years client contracts
- Improved compliance model recalls from 70% to 90%
- Reduced client bug requests from 1/day to 1/month by setting up an efficient QA process
- Learned Java to remove the Analytics dependency on Backend teams
- Developed best practices for code quality as one of the first Data Science hires and created foundational documentation that helped onboard 13 people

EDUCATION

London School of Economics

MSc Economics

- Distinction in dissertation
- Courses included Advanced Macroeconomics, Econometrics, and Quantitative Methods

London School of Economics

BSc Philosophy & Economics

- First Class Honours
- Courses included Mathematical Methods, Statistics, Econometrics, and Philosophy of Science

EVENTS & TALKS

Aug 2026

GovAI expert panel — “OpenAI x Hugging Face incident: What happened? What’s next?” ([Event](#))

Jul 2026

Secret Loyalties AI safety research hackathon — invited to serve as a judge for the hackathon organized by [Apart Research](#) and [Formation Research](#)

Jun 2026

MATS project talks at LISA and Google DeepMind London — presented the covert malicious fine-tuning project from Francis Rhys Ward’s stream to research audiences at both offices

Apr 2026

MATS 9.0 Spotlight Talk — opened the London event ([YouTube](#))

Mar 2026

FellowCast Podcast at Constellation — “Covert Malicious Fine-Tuning” ([Spotify](#))

SKILLS SUMMARY

Technology	Groovy, Java, Python, Pandas, Matplotlib, Seaborn, Scikit-learn, NumPy, PyTorch, SQLAlchemy, MySQL, HBase, Kotlin, R, Stata, AWS S3, unit testing, code review, CI/CD, Apache Spark
Data Science	Multivariate classification, SVM, logistic regression, regularization, multicollinearity, feature selection, cross-validation, class balancing, PCA, gradient boosting, t-SNE representations, KNN, and decision trees. Performance metrics: recall, precision, F-score, ROC-AUC, and PR-AUC
Statistics	Hypothesis testing, A/B testing, asymptotic theory, regression analysis, vector autoregressions, time-series analysis, panel-data analysis, fixed and random effects, and IV estimation
Software & Frameworks	Jira, TeamCity, Jenkins, GitLab, Kibana, IntelliJ, PyCharm, Bitbucket, GitHub, JupyterLab, FastAPI, and Airflow
Leadership	Team planning, mentoring, interviewing and hiring, and roadmapping
Other Interests	Oil painting, skiing, and hiking